The Extremal Principle — problem 3.7

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$(2n + 1)$ persons are placed in the plane so that their mutual distances are different. Then everybody shoots his nearest neighbor. Prove that

(a) at least one person survives ; (b) nobody is hit by more then five bullets ;

(c) the paths of the bullets do not cross ; (d) the set of segments formed by the bullet paths does not contain a closed polygon.